Latent-class and mixture-model formulations for biomarker-treatment interaction in N-of-1 trials

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2026-08-12

Contents

1 Abstract

Background. Aggregated N-of-1 clinical trials test biomarker-treatment interactions through a fixed continuous- heterogeneity regression in published practice. The psychometric literature has developed mature finite-mixture and latent-class machinery for representing discrete subpopulation structure that the biostatistical literature on N-of-1 trials has not engaged. The choice between continuous-heterogeneity and latent-class encodings of the biomarker-moderation question is a substantive scientific question rather than a parameterization convenience, but the field has not characterized when class-aware analyses become competitive with continuous regression at trial-relevant sample sizes.

Methods. We develop a five-component biomarker-moderation spectrum (covariance-only, mean-and-covariance, location-scale, heterogeneous random-effects, latent-class) bridging continuous-heterogeneity and latent-class formulations, and we catalogue the finite-mixture taxonomy (latent class analysis, growth mixture models, factor mixture models, regression mixtures) for application to within-subject longitudinal designs. We commit a heterogeneous-random-slopes generative data-generating process (DGP) and a class-aware analysis procedure (`lcmm::hlme` with gridsearch initialization), and run a 240-replicate-per-cell prototype simulation at $N = 70$, gating slope $\in \{0, 1.0, 1.5\}$, on a 4-core parallel implementation. Three test forms are recorded per replicate: the linear-mixed `bm:Dbc` Wald, the unconditional Wald on the lcmm gating coefficient, and a BIC-conditional gating procedure.

Results. In a 240-replicate-per-cell pilot we find that the unconditional Wald test on the latent-class gating coefficient inflates Type I error to 0.197 (95% CI $[0.145, 0.250]$) at the null, decisively above nominal 0.05 and consistent with the[1] and[2] overextraction predictions. The BIC-conditional gating procedure controls Type I (0.004 under the null) but is severely under-powered at trial-relevant N (rejection rate 0.025 at gating slope 1.0, 0.004 at 1.5). The linear-mixed `bm:Dbc` Wald test holds nominal Type I and reaches 0.542 power at gating slope 1.5, dominating both class-aware tests. The Monte Carlo standard error (MCSE) on power is approximately 0.030 across the grid. These pilot estimates precede the production five-study program described in the companion simulation plan, which has not yet been executed.

Conclusions. Within the 240-replicate pilot, at trial- relevant sample sizes ($N = 70$, eight measurement timepoints), neither the unconditional Wald-on-gating nor the BIC-conditional gating test is competitive with the linear-mixed baseline under the heterogeneous-random-slopes DGP. The latent-class encoding appears to require a regime of large class separation for class-aware tests to gain power; the pre-registered production program (five studies, $N \in \{35, 70, 100\}$, wider gating- slope grid) has not yet been executed and is required to map the regime where the BIC-conditional procedure becomes competitive.

2 Introduction

2.1 The aggregated N-of-1 trial and the biomarker-moderation question

The aggregated N-of-1 trial design is a class of clinical trials in which each enrolled participant cycles between active treatment and control across multiple within-person blocks, with within-person comparisons pooled across participants for population-level inference[3–6]. The design preserves the within-subject contrast that gives single-participant N-of-1 trials their inferential strength while accumulating the statistical efficiency of a multi-participant trial. For chronic conditions whose treatment response is reversible on a useful timescale, the design yields more information per participant per unit time than a parallel-group randomized trial of equivalent total sample size[7]. It has been deployed in chronic pain[8], inflammatory bowel disease, neuropsychiatric symptoms in post-traumatic stress disorder[9], and a growing portfolio of rare and orphan disease applications where parallel-group trials are infeasible[5,10].

Within this design class, a question of increasing scientific weight is whether treatment response is moderated by a measured pre-treatment biomarker. A biomarker that predicts which participants respond to which treatments would allow clinicians to stratify treatment decisions at the point of care, and the formal framework for such inference is laid out in the PATH statement[11]. The corresponding statistical estimand is the biomarker-treatment interaction term in the within-participant outcome model. Power to detect this interaction depends on the magnitude of moderation, the precision of the within-participant contrasts, the trial design (open-label, crossover, hybrid), and any time-dependent leakage between treatment phases, conventionally termed 'carryover', that erodes the cleanness of the within-subject contrasts.

Recent methodological work has documented substantial power losses for biomarker-treatment interaction inference under realistic carryover, with the loss varying sharply by how the moderation effect is represented in the data-generating process[9]. The present paper takes up a question that the prior work raised but did not resolve: how should biomarker moderation be represented in the DGP itself? Two formal answers, drawn from a long history of work outside biostatistics, dominate the available options. We begin by laying out these two answers and the methodological tradition behind each, before turning to the specific consequences for N-of-1 trial design and analysis.

2.2 Two formal answers: continuous heterogeneity and latent-class structure

The first answer represents biomarker moderation as a graded continuous effect: every participant has some degree of treatment response, the response is a smooth (typically linear) function of the biomarker, and population heterogeneity is captured by a single multivariate distribution with an interaction parameter. This is the dominant tradition in clinical biostatistics, where heterogeneity of treatment effect is routinely modeled via random slopes on the treatment indicator and via interaction terms in fixed-effects regression. In its most economical form, the moderation enters the conditional mean of the outcome as $\beta_{bm:D} \cdot B \cdot D$, where B is the biomarker and D is the (possibly continuous-decay) treatment indicator.

The second answer represents biomarker moderation as a discrete subpopulation structure: participants belong to one of a small number of unobserved classes (most commonly ‘responder’ and ‘non-responder’), the classes have qualitatively distinct treatment-response curves, and the biomarker predicts class membership without determining it. The corresponding generative form is a finite mixture in which class membership is a Bernoulli or multinomial draw with mixing probabilities that depend on the biomarker, and the within-class outcome model is class-specific. The marginal moderation observed in the data emerges from the mixture rather than from a smooth conditional-mean function.

These two answers correspond to genuinely different biological hypotheses. The continuous-heterogeneity view is consistent with a mechanism in which the biomarker reflects a graded physiological substrate (receptor density, enzymatic activity, baseline trait severity) that scales drug effect across the entire population. The latent-class view is consistent with a mechanism in which qualitatively distinct phenotypes exist (a metabolising versus non-metabolising allelic variant; a present versus absent target receptor) and the biomarker is an imperfect indicator of phenotype. The two views can produce identical marginal biomarker-outcome associations at a single timepoint and diverge sharply once the longitudinal structure of an N-of-1 trial is engaged, because carryover and within-person serial dependence interact differently with continuous and class-structured heterogeneity. This is the substantive point at which methodology imported from outside biostatistics becomes useful: both traditions have long histories of formalization and identifiability analysis in their respective fields, and the corresponding tools are mature.

2.3 Related work: two methodological traditions

The continuous-heterogeneity and latent-class answers each arrive with a distinct history that this section summarizes; the full literature review, tracing both traditions decade by decade, is given in the companion supplement (`supplement.pdf`, Section 1).

The continuous-trait view originates with Spearman's two-factor theory of intelligence[12] and was elaborated through the modern factor-analytic tradition[13,14], in which between-person heterogeneity in a latent construct is by assumption continuous. The discrete-class view emerged from Lazarsfeld's latent structure analysis[15,16], placed on rigorous footing by Goodman[17] and unified in the finite-mixture monograph of McLachlan and Peel[18]. Bartholomew[19] framed the tension between the two views as a choice between 'graded' and 'unfolding' models that the data themselves rarely settle without strong prior assumptions.

A largely independent strand developed in econometrics and marketing. Heckman and Singer[20] showed that structural parameter estimates in duration models depend sensitively on the assumed heterogeneity distribution and proposed estimating that distribution as a discrete mixture, and Lindsay[21] placed the resulting theory of mixture likelihoods on a unified footing. In marketing science, DeSarbo and Cron[22], Wedel and DeSarbo[23], and Jedidi and colleagues[24] developed covariate-dependent gating functions for segment membership, a structure formalized independently in machine learning as 'mixtures of experts'[25]. This gating-function machinery is the exact structure the biomarker-moderation problem requires: whether a single observed covariate (the biomarker) predicts latent class membership, and whether the within-class treatment coefficient differs across classes.

By the late 1990s both strands converged on hybrid factor mixture models[2,26,27] and their longitudinal extension, the growth mixture model[28–32]. Bauer and Curran[1], Bauer[33], Lubke and Muthen[2], and Nylund and colleagues[34] subsequently documented overextraction hazards: growth and factor mixture models fitted to single-component, non-gaussian data can spuriously identify multiple latent classes, and reliably distinguishing genuine from spurious class structure typically requires several hundred to several thousand participants. This caution is central to the identifiability argument developed below.

Mixture methods entered longitudinal biostatistics through Verbeke and Lesaffre[35] and Zhang and Davidian[36], and through the `lcmm`[37] and `flexmix`[38,39] R packages used in the present paper's pilot. Despite this translation, mixture and latent-class methods have not, to the best of our knowledge, been systematically applied to aggregated N-of-1 trial designs: the published N-of-1 methods literature specifies biomarker moderation as a continuous interaction term throughout[5,6,9,10,40–42], with no parallel treatment of the latent-class formulation. This gap is what the present paper addresses.

2.4 Why this matters for N-of-1 designs specifically

Three features of the aggregated N-of-1 design make the continuous-versus-discrete moderation question particularly consequential for design and analysis decisions.

First, the within-participant longitudinal structure of an N-of-1 trial provides repeated measurements at multiple drug states (on-drug, off-drug, in carryover). Under a continuous-heterogeneity DGP, the moderation signal is carried by how the within-participant treatment contrast varies with the biomarker. Under a latent-class DGP, the moderation signal is carried by class membership, which is fixed across all measurements within a participant and is therefore in principle recoverable from the full trajectory even when individual on-drug contrasts are noisy. The two formulations imply different efficient analysis strategies and different power profiles under carryover[9].

Second, N-of-1 trials are typically conducted at sample sizes of $N \in [30, 150]$, well below the sample sizes at which mixture-model identifiability has been validated in the psychometric literature. The identifiability cautions documented by[1] and[2] therefore apply with full force, and the analysis-strategy implications are not trivial: even if a latent-class DGP is the biologically correct generative model, mixture analysis at trial-relevant sample sizes may be either underpowered or unidentified, and a misspecified single-component analysis may be the more reliable inferential tool despite its bias. Resolving this trade-off is in part an empirical question that the existing psychometric simulation literature has not addressed at N-of-1-relevant sample sizes.

Third, carryover, which enters the analysis as a continuous decay in the effective drug indicator $D_{bc,it} = \exp(-\lambda t_{sd,it})$, interacts differently with the two moderation encodings. In the continuous-heterogeneity formulation, carryover erodes the moderation signal directly because the moderation enters as a product with $D_{bc,it}$. In the latent-class formulation, carryover erodes the within-class drug contrast but does not directly attenuate class membership; if class membership can be recovered from the full trajectory, some moderation signal can survive carryover that would be lost under the continuous-heterogeneity analysis. The two formulations therefore predict different power-loss profiles under carryover, and this difference is empirically large in the simulation literature accumulated by the companion carryover-sensitivity work.

2.5 Aims and structure of the paper

This paper has four aims. The first is to formalize the latent-class and finite-mixture formulations of biomarker-treatment interaction in aggregated N-of-1 trials, with explicit identifiability conditions specific to the trial-relevant sample-size regime. The second is to characterize estimation strategies (expectation-maximization, Bayesian samplers, candidate R packages) and the diagnostics required to distinguish genuine from spurious class structure at small N. The third is to report a factorial simulation study that benchmarks mixture-DGP and continuous-DGP power against the standard linear-mixed-model analysis and a class-aware analysis under realistic carryover. The fourth is to apply the developed methods to the prazosin-PTSD biomarker dataset of[9] and to compare conclusions with those obtained under the continuous-moderation analysis reported in the original paper.

The remainder of the paper is organized as follows. We begin in the notation section with a glossary that fixes terminology used throughout. The latent-class generative form, its MVN second-moment approximation, and the regimes in which the approximation breaks down are developed in three subsequent sections. The implications for power under carryover and the identifiability constraints that bound mixture analysis at trial-relevant sample sizes are addressed next. A finite-mixture taxonomy section locates the relevant model families (LCA, GMM, FMM, regression mixtures) and a biomarker-moderation spectrum section organizes the available DGP encodings moment by moment; both are given in condensed form here, with the full development in the companion supplement. Implications for analysis and design, and software for mixture analyses, are then drawn together. The discussion section addresses three open methodological questions and reports a 240-replicate pilot from the five-study simulation program described in the companion simulation plan, which constitutes the empirical core of the present line of work.

3 Notation

Throughout, B_i denotes the pre-treatment biomarker value for participant i. BR_{it} denotes the biological-response component of outcome for participant i at time t, decomposed in the Hendrickson framework into three latent response factors: BR_{it} itself, a time-variant component TV_{it}, and a placebo-belief component PB_{it}. $D_{bc,it}$ denotes drug state at time t in the analysis-side continuous-decay form $D_{bc,it} = \exp(-\lambda t_{sd,it})$ during off-drug periods and $D_{bc,it} = 1$ during on-drug periods, with $t_{sd,it}$ the time since drug discontinuation and λ the decay rate. Following the convention used throughout this series, D_{it} is reserved for the binary on-drug state and $D_{bc,it}$ for the continuous exposure-decayed indicator used here; the latter is the `Dbc` regressor of the analysis model. Z_i denotes an unobserved class label. The half-life of carryover is $t_{1/2} = \ln 2/\lambda$ and is allowed to differ between the DGP and the analysis model.

The principal psychometric and econometric terms used below are:

- **Latent class.** An unobserved categorical class, equivalent in trial vocabulary to an unobserved responder/non-responder class.
- **Latent factor.** An unobserved continuous trait governing covariation among observed variables, equivalent to a frailty (the survival-analysis term for an unobserved multiplicative random effect) or to a random intercept in a linear mixed model.
- **Class-conditional distribution f_z.** The outcome distribution within a single latent class.
- **Mixing proportion π.** The prevalence of a given latent class in the population.
- **Measurement invariance.** Whether a model applies identically across subpopulations.
- **Growth mixture model (GMM).** A longitudinal mixed model in which the population of random effects is itself a finite mixture.
- **Factor mixture model (FMM).** A hybrid model with discrete classes at the upper level and within-class continuous heterogeneity at the lower level.

- **Regression mixture / mixture of experts.** A finite mixture in which the gating function (class-membership probability) depends on covariates and within-class regressions are class-specific.
- **EM algorithm.** Expectation-maximization (EM) applied to recover unobserved class labels jointly with class-conditional model parameters.

4 Faithful latent-class data-generating processes

If a candidate predictive biomarker indexes an unobserved responder/non-responder dichotomy, the mechanistically faithful data-generating process is a finite mixture:

$$Z_i \mid B_i \sim \text{Bernoulli}(\pi(B_i)), \qquad BR_{it} \mid Z_i = z \sim f_z(t, D_{bc,it}),$$

where Z_i is the latent class indicator, $\pi(B_i) = \Pr(Z_i = 1 \mid B_i)$ is the class-membership probability (typically a logistic function of the biomarker), and the class-conditional response distributions f_0 and f_1 differ in their drug-response parameters: peak response, onset rate, or both. Within-class temporal structure follows the same modGompertz form used in single-class formulations, with class-specific parameters substituted.

We note two consequences of this DGP form that are central to what follows. First, the class label Z_i is fixed across all measurements within a participant; it is invariant to drug state and to time. Second, the class-conditional mean trajectory does depend on $D_{bc,it}$: under a full carryover off-drug period ($D_{bc,it} \to 0$) the two class-conditional means converge if the responder/non-responder distinction is purely in drug response. Class-label invariance therefore preserves participant-level information about the moderation signal that an analysis with access to Z_i could exploit, even where instantaneous between-class mean separation in BR is fully attenuated. The MVN-covariance encoding of biomarker moderation, in contrast, has no such latent-label information to exploit: the covariance is the entire signal, and its erosion under carryover is direct.

5 MVN approximation to mixture-induced moderation

A two-component mixture with class-dependent means and common within-class covariance produces a marginal joint distribution of (B_i, BR_{it}) whose first two moments are reproduced, to leading order, by a single multivariate normal with $\mathrm{Cor}(B, BR) = c_{bm}$. Under within-class independence between B_i and BR_{it},

$$c_{bm}^2 \;=\; \frac{\pi(1-\pi)(\mu_B^1 - \mu_B^0)^2}{\mathrm{Var}(B)} \cdot \frac{\pi(1-\pi)(\mu_{BR}^1 - \mu_{BR}^0)^2}{\mathrm{Var}(BR)},$$

so that c_{bm}^2 is the product of the between-class variance fractions in B and in BR. The marginal correlation that an MVN can reproduce is therefore bounded by the geometric mean of the two intra-class correlations; if either class separation is small, the achievable marginal c_{bm} is small even when class structure is otherwise well-defined.

Under mild regularity conditions (adequate class overlap and mixing proportions π bounded away from 0 and 1), the mixture's second-moment structure is well-approximated by a single MVN with differential correlation across drug states: elevated correlation where class-conditional means separate (on-drug) and shrinkage towards zero where they coincide (off-drug after carryover has decayed). For power calculations driven by t-tests on $\beta_{bm:D}$ in a linear mixed-effects model, this approximation is adequate within the regularity conditions and captures the covariance consequences of latent-class structure without committing to a discrete generative form.

6 Failure regimes of the MVN approximation

The MVN approximation is misleading in three identifiable regimes, each with a distinct empirical signature.

- **Bimodal responses.** When f_0 and f_1 are well-separated (non-responders show near-zero drug effect and responders show a large effect), the marginal distribution of BR is bimodal: a histogram of on-drug response would show two peaks rather than a single peak with gaussian scatter around it. No MVN matches the tails of a bimodal distribution, and power under the true mixture DGP will exceed the MVN-LMM prediction because a well-specified mixture analysis can exploit the bimodality. The pattern occurs in clinical settings where a subpopulation genuinely does not respond to a treatment, as in certain targeted oncology indications.

- **Strong class-membership gradient.** When $\pi(B_i)$ is steep, approaching a step function at a biomarker threshold, the biomarker behaves nearly deterministically as a class label. A thresholded mean-moderation specification (a thresholded Architecture-A variant) then becomes the better approximation, because the biomarker-response relationship is dominated by its mean-structure component. The corresponding clinical scenario is a predictive biomarker with near-perfect sensitivity and specificity for the responsive phenotype.
- **Class-varying covariance.** If autocorrelation, residual variance, or placebo-response parameters differ between classes, no single-component MVN can represent the generative structure. Analysis models assuming constant covariance are misspecified regardless of whether they represent the biomarker interaction in the mean or in the covariance. This is the latent-class analogue of heteroscedastic residual variance addressed in standard mixed- effects practice with `varIdent` or `weights` specifications, except that the stratifying variable is unobserved.

7 Implications for power under carryover

The substantial covariance-only power loss under carryover (40–60% across designs in[9]) reflects erosion of a second-moment signal. A mixture-model analysis fitted to mixture-DGP data can in principle recover additional power by estimating class membership directly, because the class label is carryover-invariant. The gap between the covariance-only carryover-sensitive power estimate and the mixture-faithful power estimate upper-bounds what a mixture analysis could recover; whether this bound is tight in the trial-relevant parameter regime is an empirical question not addressed by existing simulation infrastructure. The bound is asymptotic: at finite N, finite-sample bias can drive the realized gap below the bound or, in adverse identifiability regimes, reverse its sign.

A focused factorial simulation isolating second-moment approximation error from analysis-model mis-specification is described in the companion simulation plan (Study 1: mixture-vs-MVN factorial). The factorial crosses two DGPs (mixture and MVN) with two analyses (linear mixed model and joint latent-class mixed model). The mixture-DGP/MVN-analysis versus mixture-DGP/mixture-analysis contrast quantifies power recoverable by a faithful analysis model; the difference between the two MVN-analysis cells quantifies approximation error. A null check on the MVN-DGP/mixture-analysis cell verifies the mixture model does not extract spurious power on data with no latent class structure.

8 Identifiability at trial-relevant sample sizes

Mixture modeling is attractive on faithfulness grounds but carries identifiability costs that are acute in N-of-1 settings. The EM algorithm for finite mixtures of mixed-effects models requires either informative priors on class-membership probabilities or strong separation between class-conditional response curves; neither is assured in a trial designed to detect a moderation effect whose existence is itself uncertain. The standard diagnostic for mixture identifiability, the entropy of posterior class-membership probabilities, requires substantial sample sizes (N several hundred to several thousand) to stabilize at values that support confident class extraction in published psychometric applications[1,33,34]. Aggregated N-of-1 trials of the size contemplated in current biomarker applications, $N \in [30, 150]$, are well below this range.

The methodological implication is that mixture analysis at trial-relevant sample sizes may be either underpowered or unidentified, even when the underlying DGP is genuinely mixture-structured. The question is therefore not whether mixture formulations are more faithful, but whether the fidelity gain is recoverable at realistic sample sizes, or whether analysis-strategy mitigations available within standard mixed-effects practice (restricted analysis, weighting, within-subject contrasts; see[9]) deliver a better power-per-complexity trade. These mitigations have better-understood inferential properties at the relevant sample sizes. Resolving the mixture-versus-mitigation question empirically requires the simulation program described in the companion simulation plan; Study 5 in particular addresses small-N identifiability and Type I error explicitly.

9 Finite-mixture taxonomy

Four families of finite mixture model from the psychometric and econometric literatures bear directly on the biomarker-treatment interaction problem. Table 1 summarizes each family's structure, origin, and relevance to the N-of-1 setting; the full development of each, including the reasoning behind the package recommendations, is given in the companion supplement (Section 2).

Table 1: Finite-mixture model families relevant to the biomarker-moderation problem.

Family	Structure	Origin	Relevance to N-of-1 trials
Latent class / latent profile analysis (LCA/LPA)	Cross-sectional mixture; class-specific marginal distributions	[16;17;18]	Baseline responder-class discovery from biomarker panels; `poLCA`
Growth mixture model (GMM)	Longitudinal mixed model with a class-mixture random-effects population; class label is drug-state-invariant	[28;31]	Closest analogue of the faithful latent-class DGP in Section 3; overextraction cautions[1] apply with full force at N-of-1 sample sizes

Family	Structure	Origin	Relevance to N-of-1 trials
Factor mixture model (FMM)	Discrete classes at the upper level, continuous within-class factor structure at the lower level	[26;27;2]	Biomarker as both a noisy class indicator and a continuous within-class modulator; higher estimation cost
Regression mixture / mixture of experts	Class-specific regression surfaces with a covariate-dependent gating function	[22;25]	Biomarker governs both class probability and within-class drug-effect magnitude; `flexmix`

GMM is the closest psychometric analogue of the faithful latent-class generative form developed in Section 3: each participant belongs to a latent class with its own drug-response trajectory, and a GMM that uses the biomarker as a class-membership predictor is essentially that generative model written in psychometric vocabulary. This is why the overextraction cautions from the GMM literature[1,33] carry directly into the identifiability discussion below.

10 The biomarker-moderation spectrum

The covariance-only formulation of[9] specifies a biomarker-dependent covariance structure with population means held constant across biomarker values. Locating this formulation within the broader spectrum of biomarker-moderated DGPs is useful because the moment-by-moment organization of the spectrum (biomarker moderating the mean, the covariance, both, or higher moments) clarifies which features of the carryover-induced power loss are properties of the specification itself rather than of the underlying scientific question.

Table 2 organizes the five points on the spectrum moment by moment; the full development of each, including the measurement-invariance parallel for the covariance-only case and the GAMLSS distributional-regression formalism, is given in the companion supplement (Section 3).

Table 2: The biomarker-moderation spectrum, organized by which moment of the response distribution carries the biomarker signal.

Point on spectrum	What varies by class/biomarker	Behaviour under carryover
Covariance-only[9]	$\mathrm{Cov}(B, BR)$ only; means constant	Signal is entirely in the eroded channel; power loss is largest
Combined mean and covariance[27]	Both $E[BR \mid B]$ and $\mathrm{Var}[BR \mid B]$	Mean channel survives carryover; power loss intermediate
Location-scale (GAMLSS)[43,44]	Location and scale of the marginal distribution, no discrete classes	Analogue of the combined case without committing to a class count
Heterogeneous random-effects[35,37]	Class-specific random-intercept/slope distributions	Class label is carryover-invariant; used for the Study 5 pilot below
Pure mean moderation (Architecture A)	$E[BR \mid B]$ only; covariance constant	Signal survives carryover essentially undiminished

The covariance-only specification of[9] is the most restrictive point on this spectrum: it holds $E[BR_{it}]$ fixed across biomarker strata and places the entire interaction in $\mathrm{Cov}(B_i, BR_{it} \mid D_{bc,it})$, which is an unusual restriction within the psychometric mixture tradition (most standard FMM and GMM parameterizations permit class-specific means at a minimum). Its appeal is tractability for power simulation under multivariate normality; its cost is that it does not correspond cleanly to a natural generative mechanism, and, because the entire signal sits in a channel that carryover erodes by construction, a sensitivity analysis based solely on this point risks overstating the carryover-induced power loss a biomarker-guided trial should actually expect. The heterogeneous random-effects row is the point used for the Study 5 pilot reported below, because it preserves compatibility with the existing `nlme::lme` analysis machinery while admitting a biologically faithful latent-class generative structure at the DGP level; the `lcmm` package[37] implements this family directly.

11 Implications for analysis and design

Three implications follow from locating the covariance-only encoding within the broader spectrum.

First, the covariance-only parameterization is the most restrictive operational case of a much richer family. The substantial power loss under carryover documented in[9] is a specific property of this restrictive case and need not generalize to the mean-plus-covariance or heterogeneous-random-effects variants that are standard in psychometric and biostatistical practice. A sensitivity analysis based solely on the covariance-only DGP risks overstating the carryover-induced power loss that a biomarker-guided aggregated N-of-1 trial should expect, because the covariance-only DGP places the entire biomarker-outcome signal in a channel that carryover erodes by construction. If a portion of the signal flows through the mean structure, as in pure mean moderation or in the combined specification of[27], that component survives carryover and overall power loss is correspondingly attenuated. A complete sensitivity analysis should report power under at least three points on the spectrum: pure mean moderation, pure covariance moderation, and a combined specification.

Second, the identifiability concerns documented in the GMM and FMM literatures[1,2,33,34] bound what can be asked of mixture-based analysis models at N-of-1 trial sample sizes. Published FMM applications typically involve several hundred to several thousand participants; biomarker- trial applications with $N \in [30, 150]$ are an order of magnitude smaller. Analysis plans built around fitting `lcmm` or `flexmix` mixture models directly to trial data are unlikely to yield reliable class-membership inference at trial-relevant sample sizes and should not be relied on as the primary analysis. Mixture analysis nonetheless retains value as a sensitivity check (does class-structured residual pattern emerge from the pre-specified analysis?)

and as a DGP for power simulation (does the primary analysis remain robust to class-structured departures from gaussianity?).

Third, if N-of-1 simulation infrastructure pursues a richer DGP in subsequent work, the natural target is the heterogeneous random-slopes form rather than a full FMM. The random-slopes parameterization preserves the analysis model's compatibility with the existing `nlme::lme` machinery (the analysis model continues to be fitted as a single linear mixed model with participant-specific random treatment slopes) while admitting the biologically faithful latent-class generative structure at the DGP level. The simulation DGP is extended in a direction internal to the standard mixed-effects framework, and the analysis model remains the pre-specified linear mixed-effects fit. This preserves the audit chain of the existing framework and permits direct comparison against the covariance-only baseline under matched effect-size tuning.

12 Software for mixture analyses

R packages relevant to estimating the models surveyed above, in approximate order of fit to the longitudinal aggregated N-of-1 use case, are as follows.

- `lcmm`[37]. Joint latent-class mixed models for longitudinal continuous and ordinal outcomes; the closest available tooling for heterogeneous random-effects specification. Syntax is closely related to `nlme::lme` with a latent-class extension.
- `flexmix`[38,39]. General regression- mixture framework with user-definable component models; suitable for regression-mixture and mixture-of-experts specifications. More general than `lcmm` but requires more configuration.
- `mclust`[45]. Gaussian finite mixture models with a range of class-specific covariance parameterizations; supports[27] mean-plus-covariance specifications non- longitudinally. Useful for cross-sectional latent-profile analyses of baseline biomarker panels but does not natively handle repeated-measures structure.
- `gamlss`[43]. Distributional regression with covariate-dependent location, scale, and shape parameters.
- `poLCA`[46]. Polytomous latent class analysis for the categorical-

indicator case; useful if baseline class indicators (e.g., symptom-checklist items) are incorporated into class assignment.

- `OpenMx` and `lavaan` with mixture extensions. Structural- equation mixture modeling for FMM-style specifications; more general but with steeper configuration cost than `lcmm` or `flexmix`.

The reference implementation outside R is Mplus[47], which remains the standard environment for GMM, FMM, and related mixture-SEM specifications. Much of the published psychometric simulation literature on mixture-model identifiability and overextraction hazards[1,2,34] uses Mplus, and direct comparison with that literature occasionally requires access to it.

13 Discussion

We begin by noting that the biomarker-treatment moderation problem in aggregated N-of-1 trials sits at the intersection of two methodological traditions that have, until now, only weakly engaged with each other. The biostatistical tradition has developed efficient estimators for within-participant treatment contrasts and the meta-analytic machinery to combine them, but has typically specified biomarker moderation as a continuous interaction in fixed-effects regression. The psychometric tradition has developed mature finite-mixture and latent-class machinery for representing discrete subpopulation structure but has applied it primarily to cross-sectional or observational settings with sample sizes well above the N-of-1 range. The argument of this paper is that the choice between continuous-heterogeneity and latent-class encodings is a substantive scientific question, not a parameterization convenience, and that recognizing this choice opens both a sensitivity-analysis program (the spectrum of biomarker-moderation specifications surveyed above) and an analysis-strategy program (class-aware mixture analyses where identifiability permits, and analysis-strategy mitigations where it does not).

Unfortunately, two limitations bound what the paper currently provides. First, the simulation program that would empirically resolve the mixture-versus-mitigation efficiency question at trial-relevant sample sizes is described in the companion simulation plan but not yet executed. The five-study factorial outlined there (mixture-vs-MVN, the mean-covariance-combined triptych, heterogeneous random slopes, the three-regime breakdown grid, and the small-N identifiability and Type I error pre-flight) constitutes the empirical core of the present line of work and is the next deliverable in the program. Second, the application to real biomarker-guided trial data has not been carried out. The prazosin-PTSD analysis of[9] provides the most immediate target; the simulation program should run in parallel with re-analysis of that dataset under at least three biomarker-moderation encodings and a class-aware analysis where convergence permits.

Three open methodological questions remain. The first is whether class-aware analyses are usable as primary analyses at any realistic N-of-1 sample size, or only as sensitivity checks. The identifiability evidence from the broader mixture-model literature suggests the latter; only direct simulation in the trial-relevant parameter regime can settle this. The second is how analysis half-life should be specified when the DGP is a mixture rather than a single MVN. The continuous-decay drug indicator $D_{bc,it} = \exp(-\lambda t_{sd,it})$ has a clean interpretation under a single-component generative model; its interpretation when the underlying generative process is class-structured and class-conditional carryover dynamics differ between classes is less obvious and may require analysis-side mixture extension. The third is whether the trial design itself should respond to the moderation-encoding question: if a latent-class generative form is biologically more plausible than a continuous one, designs that maximize the per-participant contrast and minimize carryover may gain power that the standard hybrid design surrenders.

The five-study simulation program described in the companion plan is structured to address each of these questions sequentially. The factorial designs are pre-registered through ADEMP[48] tables committed before production runs; the class-aware analyses are wrapped uniformly so that simulation results are directly comparable across studies; and the infrastructure required to add the necessary DGPs to the existing simulation framework is enumerated, scoped, and sequenced to support both this paper's revisions and the companion carryover and biomarker-interaction papers in the wider research program.

Two secondary theoretical points, developed fully in the companion supplement (Section 4), are worth flagging briefly here. First, the analysis model used throughout this program is a four-term linear mixed model rather than a component-matched non-linear model that mirrors the DGP's biological-response, placebo-belief, and time-variant decomposition; the supplement gives four reasons this simpler analysis is the right choice at trial-relevant N, and describes a phase-indicator extension for papers where pharmacological-versus-placebo disentanglement is a secondary scientific question. Second, the latent-class DGP specified for Study 1 is a natural mechanism for generating apparent subadditivity (a non-zero $BR \times PB$ interaction coefficient) in a downstream single-component analysis, even absent any direct mechanistic interaction; the Study 1 pre-registration includes a `pb_class_correlation` axis designed to let the analyst distinguish this emergent mechanism from genuine mechanistic interaction empirically.

13.1 Progress to date

The five-study production program is pre-registered under the ADEMP framework of Morris, White, and Crowther[48] at `analysis/scripts/latent-class-mixture-application/00-ademp-pre-registration.md`; the present manuscript reports a Study 5 pilot at 240 replicates per cell.

Phase 1 of the simulation program has reached the infrastructure-and-pilot milestone. Three components are committed and exercised end-to-end on the host R session. ADEMP tables for all five studies[48] are committed at `analysis/scripts/latent-class-mixture-application/00-ademp-pre-registration.md`, with cell counts (approximately 2,300 cells across the program), seeded random-number control, an effect-size tuning sub-study, and a deviation-log specification. Subsequent production runs are diagnosed against these pre-registered estimands rather than against any post-hoc wrapper choice.

Class-aware analysis wrapper. An `lcmm_analysis()` wrapper around `lcmm::hlme` is committed at `01-lcmm-wrapper.R` with the same input-output contract as the existing `lme_analysis()`. The wrapper uses a two-stage fit (a single-class `hlme` is fitted first to anchor starting values for a `gridsearch`-based multi-start `hlme(ng=2)` fit) and extracts two moderation signals: the gating-coefficient z-test on the biomarker (the primary class-aware moderation test, identifying whether the biomarker predicts class membership) and the within-class $\beta_{bm:D}$ t-test (the secondary signal, retained for inferential symmetry with the linear-mixed baseline).

Heterogeneous-random-slopes DGP. A wrapper around `generateData()` that adds a Bernoulli class-membership channel with logistic gating in the standardized biomarker and class-specific multipliers on the BR factor at on-drug timepoints is committed at `02-heterogeneous-slopes-dgp.R`. On a single-replicate smoke test ($N = 70$, gating slope 1.5, responder multiplier 1.0, non-responder multiplier 0.0) the DGP produces 46% responders and a marginal $\mathrm{Cor}(B, BR) = 0.63$ that the linear-mixed `bm:Dbc` test detects ($p = 0.002$) without class awareness; the `lcmm` fit returns a posterior class-assignment entropy of 0.86 and a BIC 170 lower than the matched single-component MVN-DGP fit, indicating that the mixture model discriminates the two DGPs at this parameter cell.

Study 5 pre-flight pilot. A 100-replicate pilot under the single-component null DGP (the covariance-moderation architecture with $c_{bm} = 0$, $N = 70$, hybrid open-label (OL) design) is reported at `output/03-study5-pilot.rds`. Per-replicate timing is 0.05 s for the linear-mixed baseline and 1.57 s for the class-aware analysis (32:1 cost ratio, consistent with the smoke-test estimate). `lcmm` convergence rate is 96%; the four non-converging replicates encountered numerical issues during the gridsearch initialization.

Type I error at $\alpha = 0.05$ (MCSE 0.022) varies sharply across candidate test forms, motivating the choice of the primary class-aware moderation test:

Test	Rejection rate
lme `bm:Dbc` Wald t-test	0.000
lcmm gating-on-`bm` Wald z (unconditional)	**0.250**
lcmm within-class `bm:Dbc` Wald z	0.000
BIC selects ng=2 over ng=1	0.010
LRT chi-sq(df=4), naive reference	0.146
BIC-conditional gating procedure	0.000

The unconditional Wald-on-gating test rejects at five times the nominal rate. Diagnosis of the rejecting replicates shows that the empirical standard deviation of the gating coefficient across replicates (sd = 63.9) is roughly fifty times the median within-replicate Wald SE (0.86), and the empirical sd of the z-statistic is 1.47 against a nominal 1.0. The gating coefficient is centered at zero (median -0.10, sign balance 46/54) and shows no systematic bias; the inflation is driven by the conditional Wald SE under-estimating the true variability of $\hat{\beta}_{\text{gating}}$ at trial-relevant N. This is a finite-sample identifiability artifact predicted by the[1] and[2] overextraction literature: when the data have no class structure but `hlme` is forced to fit `ng=2`, the optimizer latches onto random patterns to define a class boundary, and the resulting Wald SE does not absorb the labeling uncertainty.

The naive LRT against a chi-squared($df = 4$) reference is similarly anti-conservative (rejection 0.146), because the true LRT distribution at the `ng=1` boundary is a chi-bar-squared mixture rather than a chi-squared ([18], ch. 6). The bootstrapped LRT of[34] is the standard calibration but is computationally expensive at the per-fit budget of a class-aware simulation.

The pragmatic class-aware moderation test is therefore a BIC-conditional gating procedure: fit `ng=1` and `ng=2`, prefer `ng=2` only if BIC supports it ($\Delta_{\text{BIC}} > 0$, or more conservatively > 6 per Raftery 1995), and only then test the gating coefficient. At $N = 70$ under the null DGP, BIC selects `ng=2` in 1% of replicates, and zero of those replicates also reject the gating Wald, giving the conditional procedure a Type I error of 0 at the pilot's resolution. The conditional procedure is therefore usable as a primary moderation test at trial-relevant N. Whether the same conditional procedure has adequate power to detect a true two-class DGP is the central question for the rest of Study 5; pilots at non-null cells will measure this directly.

The diagnostic distinction matters for the paper's argument. The earlier proposition that 'class-aware analyses are not usable as primary at trial-relevant N' is too strong as stated: it applies to the unconditional Wald-on-gating test but not to the BIC-conditional procedure. The corrected proposition is that the unconditional Wald test on a single mixture-model coefficient is unreliable at small N and requires a model-comparison wrapper (BIC or BLRT) to control Type I error.

Production-budget implication. Extrapolated to the pre-registered cell counts (Study 1 has 768 cells × 1,000 replicates = 768,000 fits per analysis arm), the class-aware arm of Study 1 is approximately 340 hours of serial CPU time at the pilot's mean fit time. Parallelisation across 8–16 cores reduces this to 22–44 hours per arm. Studies 2 and 3 require similar budgets; the Study 5 production run (5,000 replicates × three sample sizes × three class counts × three DGP families) is the largest single block at approximately 60 hours per arm on 8 cores. The full program is feasible on a single workstation over a week of background compute, with the effect-size tuning sub-study and Study 5 pre-flight running first to finalise effect-size axes and identifiability cuts before committing to Studies 1, 2, and 3.

Open question raised by the pilot. With Type I error of the BIC-conditional procedure controlled at the null at $N = 70$, the next-most-pressing identifiability question is whether the BIC selection step has enough sensitivity to detect a real two-class DGP at trial-relevant N. A class-aware procedure that is conservative under the null but rarely fires under true class structure is no more useful than a procedure that fires too often. Pilot 4 (planned for the next iteration) will fit the same BIC-conditional procedure to the heterogeneous-random-slopes DGP from Study 3 (with several values of the gating slope and class-separation Δ) and report the power profile of the conditional procedure alongside that of the linear-mixed baseline. We expect that the BIC-conditional procedure will outperform the linear-mixed `bm:Dbc` test at large class separations (where `ng=2` BIC consistently wins) but will underperform it at moderate class separations (where BIC is undecided and the procedure falls back to a default that does not exploit the latent-class structure).

13.2 Pilot results (Study 5, 240 replicates)

The Study 5 prototype was rerun at 240 replicates per cell on 4-core `parallel::mclapply` with three gating-slope levels (gs $\in \{0, 1.0, 1.5\}$) and three test forms (lme `bm:Dbc` Wald, `lcmm` gating Wald unconditional, BIC-conditional gating). Wall time was 558 s; convergence was 92.9% under the null and 98.3–98.8% at non-null cells. Per-replicate output is at `analysis/data/quick-sim/03-latent-replicates.rds`; Morris ADEMP summary at `03-latent-summary.txt`. MCSE at the working power near 0.5 is approximately 0.030.

The unconditional Wald-on-gating Type I error of 0.197 (95% binomial CI $[0.145, 0.250]$) is now decisively distinguishable from the nominal 0.05 (more than five MCSE of separation) and is statistically consistent with the 100-replicate pilot's 0.250 point estimate. Power at gs = 1.0 (0.304) and gs = 1.5 (0.360) sits below the lme baseline (0.325 and 0.542 respectively), so the unconditional Wald buys nothing in exchange for the Type I inflation. The BIC-conditional procedure does control Type I (0.004 under the null, with upper CI near 0.013), but its power is 0.025 at gs = 1.0 and 0.004 at gs = 1.5: under the heterogeneous-random-slopes DGP at $N = 70$, the BIC step selects `ng=1` in essentially every replicate and the conditional procedure approximates a never-reject rule. The non-monotonicity (gs = 1.5 below gs = 1.0) suggests that as the heterogeneous-slopes signal strengthens, a single-class lme captures it well enough that BIC continues to favor `ng=1` and the conditional procedure does not fire.

The lme `bm:Dbc` test holds nominal Type I (0 of 240 under the null) and reaches 0.542 power at gs = 1.5, dominating both `lcmm` tests at this sample size and DGP regime. We therefore conclude, on the evidence of this pilot, that at trial-relevant N and under the heterogeneous-random-slopes generating mechanism examined here, neither the unconditional Wald-on-gating nor the BIC-conditional gating test is competitive with the linear-mixed baseline: the unconditional test is severely anti-conservative, and the BIC-conditional test is severely under-powered. A canonical run at 1,000 replicates per cell with a wider gating-slope grid (and at $N \in \{35, 70, 100\}$) is required to map the regime in which the BIC-conditional procedure does become competitive, plausibly at large class separations rather than under heterogeneous random slopes.

14 Conflict of interest

The authors declare no conflicts of interest.

15 Funding

This work received no specific funding.

16 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte

Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

17 Reproducibility

This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (4-core `mclapply` for the `lcmm`-bottlenecked run), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build. The class-aware analyses additionally use `lcmm` with `gridsearch` initialization.

Random-number generator. Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

Data and code availability. Per-replicate simulation outputs from the 240-replicate Study 5 pilot are checkpointed at `analysis/data/quick-sim/03-latent-replicates.rds`; Morris ADEMP cell-level summaries at `analysis/data/quick-sim/03-latent-summary.txt`; the effect-size calibration table at `analysis/data/quick-sim/03-calibration-table.rds`. Driver scripts are under `analysis/scripts/latent-class-mixture-application/`, including the pilot driver `03-study5-pilot.R` (executed) and the production driver `10-study5-production.R` (committed; the five-study production program has not yet been run). The pre-registered ADEMP plan is at `analysis/scripts/latent-class-mixture-application/00-ademp-pre-registration.md`. The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/03-latent-class-mixture-application/`.

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